Participating schools

A competition such as this would not be possible without the support of schools, and their computer science teachers, around the world. Thank you for all you do in encouraging and challenging the next generation of computer scientists.

The following 206 schools are participating in Coding Quest. Is yours?

- Alcuin Dallas, United States (5 participants)
- American British Academy Muscat, Oman (16 participants)
- American Community School, Abu Dhabi Abu Dhabi, United Arab Emirates (14 participants)
- American International School of Budapest Budapest, Hungary (19 participants)
- American Robert College Istanbul, Türkiye (53 participants)
- American School of Barcelona Barcelona, Spain (40 participants)
- American School of Milan Milan, Italy (11 participants)
- Anatolian TUBITAK Science High School Kocaeli, Türkiye (23 participants)
- Ann Arbor Huron HS Ann Arbor, United States (1 participants)
- Antelope High School Antelope, United States (1 participants)
- Arapahoe High School Centennial, CO , United States (6 participants)
- Arizona College Preparatory High School Chandler, United States (1 participants)
- Asia Pacific International School Hauula, Hawaii, United States (1 participants)
- Aurora High School Aurora, Canada (1 participants)
- Australian International School Ho Chi Minh City, Viet Nam (2 participants)
- Aves International Academy Tema, Ghana (4 participants)
- Bangkok International Preparatory & Secondary School Bangkok, Thailand (60 participants)
- Bangkok Patana School Bangkok, Thailand (24 participants)
- Basha High School Chandler AZ, United States (19 participants)
- Bavarian International School gAG Haimhausen, Germany (8 participants)
- Beauchamp City Sixth Form Leicester, United Kingdom (48 participants)
- Bentonville High School Bentonville, Arkansas, United States (1 participants)
- Bilingual Gymnasium Česká Bratislava, Slovakia (4 participants)
- BIS Hanoi Hanoi, Viet Nam (25 participants)

Top 10 schools

Calculated from the top 5 students from each school to have completed the most problems in the least time cumulatively for the event.

1	King George V 45 solved in 17:20:50.	2	12	13	8	42	20
2	Anatolian TUBITAK Science High School 45 solved in 22:15:44.	1	4	7	19	59	42
3	Sha Tin College 45 solved in 44:14:05.	12	10	7	31	8	
4	The British International School, Budapest 45 solved.	1	6				
5	Bangkok International Preparatory & Secondary School 45 solved.	1	6	3	5	19	6
6	Garden International School 45 solved.	2	3	3	3	18	13
7	Bromsgrove International School Thailand 45 solved.	3	3	1	10	9	
8	Parker/Craig High School 44 solved.	2	4	6			
9	Victoria Shanghai Academy 43 solved.	1	3	4	12	14	
10	Jewish Free School 42 solved.	1	1	2	6	2	

Top 10 individuals at Victoria Shanghai Academy

Members of your school community to have completed the most problems in the least time (student and teacher accounts will appear here).

1	Micah 9 solved in 8:14:44.	2	1	5	1	4	1	
2	Andrew Chiu 9 solved.							
3	Y07H William Han CHEN 8 solved in 5:57:30.	1	1	1	3	2	1	4
4	Neill Lee 8 solved.							
5	Anonymous 8 solved.	1	1	1	1	3	3	2
6	Jason 8 solved.	1	1	5				
7	Y11D1 Siqi (Kent) ZHANG 7 solved in 6:24:47.	1	1	2	3	3	1	
8	David SUN 4 solved.	1						
9	Y11D1 Kiu Sum (Sam) LEUNG 3 solved.							
10	Y10T Wang San (Max) LIU 3 solved.							

Top 50 students globally

Students to have completed the most problems in the least time.

1	Yuanxia Xu 9 solved in 1:46:24.	1	6	2	9	8	1		
2	Alex CLIMIE 9 solved in 2:00:42.	4	3	2	3	3	2	9	
3	Ian Yu 9 solved in 2:42:58.	5	3	3	5	6	3		
4	Kutay Demirbaş 9 solved in 4:24:14.	1	2	2	1	2	2	5	3
5	Enes Ç 9 solved in 4:56:05.	1	3	2	2	2	2	6	2
6	Andre CHUI [10C13] 9 solved in 5:16:18.	1	1	3	2	1	1	5	1
7	Miguel Fernandes 9 solved in 5:43:36.	1	1	1	7	2	5	2	
8	Adam ARIF WEBB-JOHNSON 9 solved in 6:29:22.	1	2	1	2	6	1		
9	Dhairya 9 solved in 6:43:47.	2	1	1	2	1	5	1	
10	Timothy LAM 9 solved in 6:58:04.	4	1	3	3	2	2	6	
11	Sean Huang 9 solved in 6:59:02.	1	1	2	8	1	4	3	
12	Ayşe Betül Çakır 9 solved in 7:11:59.	2	4	2					
13	Micah 9 solved in 8:14:44.	2	1	5	1	4	1		
14	cheunt3 9 solved in 9:43:00.	1	3	2					
15	Luka Miyoshi 9 solved in 12:40:28.	1	2	6	1	1	2		
16	Gavyn 9 solved in 13:12:45.	1	3	1	1	2	1	7	
17	Daiki Fujita 9 solved in 14:06:54.	1	3	1	7	2	4	1	
18	Jasper NG 9 solved in 16:24:03.	1	1	1	3	1			
19	Pip Purttiponthanee 9 solved.	6	1	1	7	1	7	1	

2024 challenge is open!

Coding quest will return in 2024 from Monday 4th March to Friday 15th March. Set your reminders now and see you then!

(Due to a technical error with the day 10 problem, that problem was deleted)

4 March 2024

Purchase tickets

5 March 2024

Broken firewall

6 March 2024

Hotel door code

7 March 2024

Closest star systems

8 March 2024

Busy moon rovers

11 March 2024

Time to play fair

12 March 2024

The purge

13 March 2024

Connecting cities

14 March 2024

Mining tunnels

Purchase tickets Day 1

The year is 2224. You have spent most of the past year on Earth, catching up with friends and family after your travels of recent years. It didn't take too long, however, for you to long for spending some more time in space. You have decided that what you need is to spend the next two weeks on holiday to an Earth orbiting Space Station and the Moon base, Luna city. Let's go on an adventure!

The first thing you need to do is purchase your rocket flight tickets.

Rocket flight companies are worse then the airlines of the 21st century when it comes to extra fees and charges. If you know the deals, you can also find numerous discounts and deductions for various loyalty programmes and so forth. Always in the hunt for a good deal, you scrape the internet searching for every possible option available so you can get the best possible price on your tickets.

You recorded all the details of different options in a file as you found them (your input data). There are 20 different space travel companies you have researched the deals of. Determine the total cost for each company.

- Anything recorded as a *seat*, *meals*, *luggage*, *fee* or *tax* must be added to the cost for that company
- Anything that is recorded as a *discount* or *rebate* can be subtracted from the cost for that company

After determining the final cost each, what is the final cost of the cheapest option?

Example

Consider the following example,

```
AAA: Seat 9997
BBB: Discount 2886
DDD: Luggage 3500
AAA: Tax 156
CCC: Fee 9468
BBB: Fee 9378
AAA: Discount 3103
DDD: Rebate 967
```

- Spaceliner AAA has a seat cost of 9997, tax of 156 and a discount of 3103. That brings AAA to a total of `9997+156−3103 = 7050`
- Spaceliner BBB has a fee of 9378, and a discount of 2886. That brings BBB to a total cost of `9378−2886 = 6492`
- Spaceliner CCC has a fee of 9468 and that's all, so it's final cost is `9468`
- Spaceliner DDD has a luggage cost of 3500 and a rebate of 967, so the final cost is `3500−967 = 2533`.

Based on the small example, space liner DDD would be the cheapest, so the answer would be `2533`.

Your task

Of the options presented, what is the final cost of the cheapest option?

Broken firewall Day 2

Tickets in hand, you board your rocket flight ready for your vacation. Just as you are settling into your seat, the pilot can be heard on the speaker system. It seems they are having some issues with the onboard network and are looking for a Computer Scientist to help them out. Keen to see if you can get a free upgrade as a reward, you volunteer your services!

It seems the network router has been accidentally reset and is blocking legitimate traffic between the rocket systems and mission control, while simultaneously also allowing the onboard passengers to overwhelm the wifi.

On inspection, you determine the router is using Internet Protocol version 4. You setup a data logger and record the IP headers of 1000 packets for analysis, your input data.

Speaking to the captain, you are advised that internal ship systems are in the 192.168.0.0 through to 192.168.254.254 IP address range, while passenger wifi is on the 10.0.0.0 to 10.0.254.254 IP address range. Any other addresses are external to the ship (for instance the ship control center, space traffic control, or just other internet locations for passenger web browsing).

To start with, you decide to determine the ratio of network traffic that originates or is destined for the ships internal systems, compared to that which originates or is destined for the passenger wifi.

Example

Consider the first 5 packets you recorded the headers for.

```
4500037700000008306f39f0A000bc1d7253441
4500007f0000000005065de1c0a800b833c555ee
450002e50000000008061ef5c0a8796698721661
4500017e00000000b206e54e88c7fd4f0A00244c
45000164000000009d06d73c0A0000b7e0b143b8
```

Knowing Internet Protocol version 4, you know that bytes 3 and 4 are a 16 bit integer for the length of the packet, bytes 13 to 16 represent the source IP address and bytes 17 to 20 represent the destination IP address.

Looking at the first packet, you identify the length as being the hexadecimal `03 77` which is the decimal number 887. Additionally you identify the source IP address as being the bytes `0A 00 0b c1` which converts to an address of `10.0.11.193`. The destination IP address has the bytes `d7 25 34 41` which converts to `215.37.52.65`.

For the first 5 packets, the length, source IP address and destination IP addresses would convert into the following:

```
4500037700000008306f39f0A000bc1d7253441 - Length 887 bytes, source IP address 10.0.11.193, destination 215.37.52.65
4500007f0000000005065de1c0a800b833c555ee - Length 127 bytes, source IP address 192.168.0.184, destination 51.197.85.238
450002e50000000008061ef5c0a8796698721661 - Length 741 bytes, source IP address 192.168.121.102, destination 152.114.22.97
4500017e00000000b206e54e88c7fd4f0A00244c - Length 382 bytes, source IP address 136.199.253.79, destination 10.0.36.76
45000164000000009d06d73c0A0000b7e0b143b8 - Length 356 bytes, source IP address 10.0.0.183, destination 224.177.67.184
```

With that information, you determine the total bytes of network traffic for the ships internal systems as being 868 bytes, and the passenger wifi as being 1625. Therefore the ratio would be `868/1625` which would be your answer (submit in that style, don't perform the division).

Your task

Analyse all 1000 packet headers, and calculate the total number of bytes for ships internal systems, and the total number of bytes for passenger wifi. Submit the ratio as your answer.

Hotel door code Day 3

You have arrived at the Space Station and make your way to the onboard hotel. After completing the check in procedure, the hotel receptionist sends your room keycode to your phone. Unfortunately it seems the file is a format that is not compatible with your phone, so you are going to have to decode it manually.

You ascertain that the file is supposed to be an image file but it seems to be compressed using a form of Run length encoding. Performing a summation of all the values in the data, you realise there are 8000 pixels worth of data, and that the data appears to just be black and white values. You hypothesize that the image would be close to a square, so with 8000 pixels you decide to try 80 rows x 100 columns first.

Example

Consider small example of run-length encoded data below that renders into a 10 wide by 6 high image.

```
12 6 4 1 9 5 5 1 9 1 7
```

By rendering the first number as `.` pixels, and the second number as `#` pixels, and then alternating between the two until completion, would create the following image.

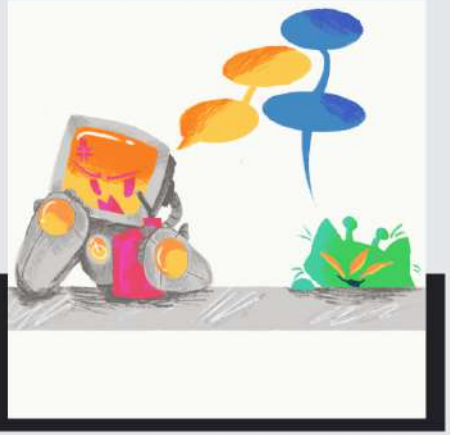
```
..........
..######..
..#......
..#####...
..#......
..#......
```

Which can be read as the letter `F`.

Your task

Find your room code by decoding the run-length encoded input data into a 80 rows x 100 columns pixel image. The answer is the number you can read in the resulting image.

Closest star systems Day 4



You are relaxing at a bar and strike up a conversation with a couple of other space travellers. You start talking about the vastness of space and how close (or far) different star systems are from each other. You actually get into a little bit of a disagreement because they are saying our Solar system and the Proxima Centauri system are the closest to travel between in the Milky Way at 4.247 light years. You aren't so sure and imagine that amongst all the star systems out there, there must be some that are closer to each other than that.

Determined to get to the truth of the matter, you download data for a List of Nearby Stars to extract the coordinates for stars within about 26 light years of Earth. When a system has multiple stars, you just take the coordinates of the first star in that system - your input data.

Looking at the first 5 systems in the data,

System	Dist	X	Y	Z
Proxima Centauri	4.247	2.945	-3.056	-0.143
Barnard's star	5.963	4.958	2.980	1.449
Luhman 16 A	6.503	1.697	-6.249	0.600
WISE J085510.83-071442.5	7.532	-3.967	-5.664	2.985
Wolf 359	7.856	-1.916	-3.938	6.522

The `Dist` column indicates the system distance from our solar system. The `X`, `Y` and `Z` columns indicate each star systems distance from sol in the x, y and z planes. To calculate the distance between any two star systems, requires finding the difference in their x/y/z distances and using those differences with Pythagoras to find the distance for that pair of star systems.

For example, to find the distance between Proxima Centauri and Barnard's star would be

```
distance = squareroot( square( x1-x2 ) + square( y1-y2 ) + square( z1-z2 ) )
         = squareroot( square( 2.945-4.958 ) + square( -3.056-2.980 ) + square( -0.143-1.449 ))
         = 6.559 lightyears
```

Example calculations for these first five systems would be:

FROM	TO	CALCULATED DISTANCE
Proxima Centauri	Barnard's star	6.559
Proxima Centauri	Luhman 16 A	3.508
Proxima Centauri	WISE J085510.83-071442.5	8.023
Proxima Centauri	Wolf 359	8.296
Barnard's star	Luhman 16 A	9.825
Barnard's star	WISE J085510.83-071442.5	12.519
Barnard's star	Wolf 359	10.993
Luhman 16 A	WISE J085510.83-071442.5	6.173
Luhman 16 A	Wolf 359	7.312
WISE J085510.83-071442.5	Wolf 359	4.438

Of the above limited data, the closest pair of systems is Proxima Centauri with Luhman 16 A at a distance of 3.508 light years. In this instance, `3.508` would be your answer.

Your task

Perform Pythagoras theorem in 3 dimensions on every pair of star coordinates in the input data and find the pair that are the closest distance from each other.

Give your answer to three decimal places.

Busy moon rovers Day 5

You make the short trip from the Space Station over to the Moon base, Luna city, for a few days. You bump into an old friend of yours who runs a logistics and delivery business there, running automated rovers to deliver supplies to the various outposts around the moon.

Your friend expresses to you they are curious how far their rovers travel per day and you offer to help out.

Your friend supplies you a spreadsheet detailing the distances between the respective outposts, and a record of the stops made by the automated rovers throughout the day (your input data). Each rover starts and ends the day at the main base so they can recharge overnight.

Example

Consider the following simplified example with the distances between base camp and 3 outposts.

```
        base    ta00    cx22    xj84
base       0   55457   63529   61302
ta00   55457       0  111890   35768
cx22   63529  111890       0   98977
xj84   61302   35768   98977       0
```

And the following example rover journey log:

```
Rover 1 route: base -> cx22 -> ta00 -> base -> xj84 -> base
```

This particular rover traveled as follows:

- From base to cx22 - a distance of 63529.
- From cx22 to ta00 - a distance of 111890.
- From ta00 to base - a distance of 55457.
- From base to xj84 - a distance of 61302.
- From xj84 to base - a distance of 61302.

Sum these distances to calculate the total distance traveled by this rover as `353480`.

Your task

Process the journey distances of all 10 rovers to calculate the total distance traveled. That is your answer value.

Time to play fair Day 6

You receive a message from your sister back on Earth. Just as a couple of years ago, you still communicate with each other using a private cipher - for a little bit of nerdy fun between siblings. Recently you both agreed to switch to using a Playfair cipher, so that's what you are going to have to figure out in order to read the message.

Since there are a few ways of implementing playfair, you remember the two of you agreed to the following stipulations:

- Replace any and all occurrences of `j` with the letter `i`.
- For words of odd length, append an `x` (unless the last letter is already `x`, in which case use a `q`).

Example

Consider this example playfair grid, created using the encryption key of `helloworld` and the phrase to decode of `wp nehslv ewgw`.

```
h e l o w
r d a b c
f g i k m
n p q s t
u v x y z
```

- Notice that repeat occurrences of letters have been ignored when created the grid (only the first `l` and `o` appear).
- The rest of the alphabet has been added, ignoring letters used by the cipher key and the letter `j`.
- Break each word of the phrase into letter pairs: `wp`, `ne hs lv` and `ew gw`.
- `wp` appears on different rows and columns in the grid. That means `w` is replaced by the letter at the same row as `w` but the column of `p`, which is the letter `e`. Likewise, `p` is replaced by the letter at the same row as `p` but the column of `w`, which is the letter `t`.
- The pairs `ne`, `hs`, `lv`, and `gw` all appear in different rows and columns to their paired letter, so the same method applies.
- The pair `ew` appear in the same row to each other, so in that case, the `e` is replaced by the letter immediately to the left which is `h`, and the `w` is also replaced with the letter immediately to the left which is `o` (wrap around if required. For instance `h` would be decoded with `w` in the above example). A similar process applies if a pair of letters appear in the same column, you decode with the letter immediately above each letter.

The above phrase decodes to `et phonex home`. The `x` in `phonex` appears as the `x` was added to ensure all words have an even number of letters. It is one of the quirks of the playfair algorithm.

Your task

Obtain your decryption key and message to decode from your input data. The decoded message is your answer.

The purge Day 7

You try to download all the photos your sister sends you but your phone produces an error message that there isn't enough space available. It seems all your constant programming and tinkering has left your phone filled with a lot of unnecessary files. For instance you see a lot of folders with names like "TEMPORARY" or "DELETE LATER".

You decide to take a few minutes to clean up the file system on your phone. Obviously you need to be careful not to delete any files your phone actually needs to operate.

Any file or folder containing the word `delete` or `temporary` are safe to delete, but just how much disk space will it free up? It's one thing to total all the files containing `delete` or `temporary`, but if a folder name contains it, then you will also be deleting all the files contained within that folder (regardless of their names), plus any sub folders within the offending folder! You could find yourself going into multiple layers of folders to total all the file sizes!

Example

Folder 0 is always the starting point of your search, and everything else is contained within it. On a Mac this might be folder `/`, or on Windows this could be your `C:/`.

Let's look at a small example...

```
Folder: 0
 - taskmgr.exe 5065932
 - Customer_Feedback_Compilation_2024.xlsx 2646384
 - VLC-3.0.16-win64.exe 3971817
 - ProductLaunch2024.png 3712336
 - temporary_573 5048816
 - delete_708 2054307
 - temporary_023 [FOLDER 1]
Folder: 1
 - Conference_Break_Music.mp3 5179931
 - SlackSetup-x64-4.3.2.exe 2384929
 - Strategic_Plans [FOLDER 2]
 - Client_Dinner_Mar_2024.jpeg 5364778
 - FileZilla_3.55.1_win64-setup.exe 4623628
 - Charity_Event_011524.HEIC 2134414
 - Office_Christmas_Party_2023.jpeg 687062
Folder: 2
 - Product_Video_Soundtrack.aiff 813896
 - Operations_Manuals [FOLDER 3]
 - Signed_NDA_JohnDoe_021523.pdf 3257437
 - delete_930 9940460
 - Client_Acquisition_Strategies [FOLDER 4]
 - temporary_493 1332303
 - Marketing_Brochure_Image1.PSD 5913702
Folder: 3
 - ProductLaunch2024.png 4396529
 - Motivational_Morning_Playlist.m3u 5619626
 - Network_Configuration_Settings.txt 1068226
 - CRM_Database_Export_021724.csv 5812973
 - Competitor_Analysis_0224.pdf 1088620
 - Employee_Training_Videos_Link.txt 267104
 - delete_530 7150742
Folder: 4
 - temporary_751 1051994
 - delete_208 6042521
 - Logo_Rebrand_Options.svg 3438585
 - Node-v14.17.3-x64.msi 2056068
 - Expense_Report_Jan_2024.pdf 5775782
 - user32.dll 2371618
 - delete_027 9003131
```

Folder 0 contains 1 file for deletion, `delete_708` of size `2054307`. Folder 0 also contains a folder called `temporary_023`. Due to the name of the folder, that means everything within it will also get deleted. To total the disk space consumed by the files in `temporary_023`, you need to go through the contents of that folder which we are told is `[FOLDER 1]`.

Jump to folder 1, and total all the files within it, plus also `[FOLDER 2]` which is inside it.

Folder 2 contains folder 3 and folder 4, so the file sizes in those also need totalling.

Converting the folder numbers to their folder names, the above structure could be visualised with a tree of the following:

```
/
+-- temporary_023
    +-- Strategic_Plans
        +-- Operations_Manuals
        +-- Client_Acquisition_Strategies
```

Once you have traversed the entire folder structure in this example, the total bytes that would be freed up are `103879262`, which would be the answer.

Be aware that as you are totalling files to be deleted, take care not to double-count some files. For instance the file `delete_027` inside folder 4 is already getting deleted due to the folder it resides in, so the file size doesn't need counting separately.

Connecting cities Day 8

While there are a couple of major cities on the moon, currently the network connectivity between two of them has so far been limited to the use of satellites in moon orbit. There are several different options being considered to provide alternative means of network connectivity between these cities, which are 856 kilometres distant from each other.

- Option 1: Single-mode fibre cabling can run for up to 40 kilometres before repeaters are required
- Option 2: Microwave radio transmission can transmit for 12 kilometres before repeaters are required
- Option 3: Laser links can transmit for 2 kilometres before repeaters are required
- Option 4: Copper cabling can transmit for 1 kilometre before repeaters are required

Each of the options has various benefits and drawbacks. Microwaves and lasers can transmit over valleys, or up and over mountains without having to install cabling in complicated terrain. Fibre optic cabling can be used for long distances, but needs to be buried to keep it safe from damage which increases the expense. Copper cabling is convenient for locations within existing towns and cities.

A friend of yours is on the planning committee for this new network connection and has asked for your advice.

Looking at the various range capacity of each link involved, you soon realise there are a lot of different options available to achieve the 856 kilometres of distance.

How many different options exist?

Example

Consider a smaller example where a 5 kilometre distance needs to be connected, and you have 3 different options available each with a range of 3 kilometres, 2 kilometres, or 1 kilometre.

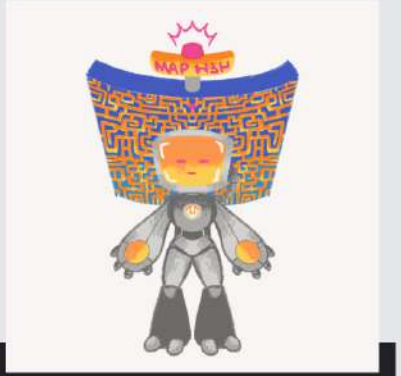
The number of options available in this smaller scenario would be 13. Each of these options is as follows...

```
3 + 2
3 + 1 + 1
2 + 3
2 + 2 + 1
2 + 1 + 2
2 + 1 + 1 + 1
1 + 3 + 1
1 + 2 + 2
1 + 2 + 1 + 1
1 + 1 + 3
1 + 1 + 2 + 1
1 + 1 + 1 + 2
1 + 1 + 1 + 1 + 1
```

Your task

To provide network connectivity across a distance of 856 kilometres, and 4 options that can each cater to distances of 40 kilometres, 12 kilometres, 2 kilometres, or 1 kilometre, how many different options are there to cover this distance?

Mining tunnels Day 9 Part 1



It's your second last day on the moon before your return to earth, so you have taken a tour of the old mining tunnels that were dug out in the early days of the moon's settlement. You've thoroughly enjoyed the tour but have become separated from the tour guide and have realised you need to find your own way out. Fortunately there is a big map on the wall nearby. It seems a lot of the tunnels have dead ends, but that there are still many different paths you could take to exit. You're starting to get a little hungry though (it is dinner time after all!) so you don't want to wander the tunnels longer than necessary.

One important issue with the tunnels is that there are tunnels on two different levels. Sometimes it is quicker to go up or down a level as part of your journey to the exit.

Process the map to determine the shortest route out of the tunnel system.

Example

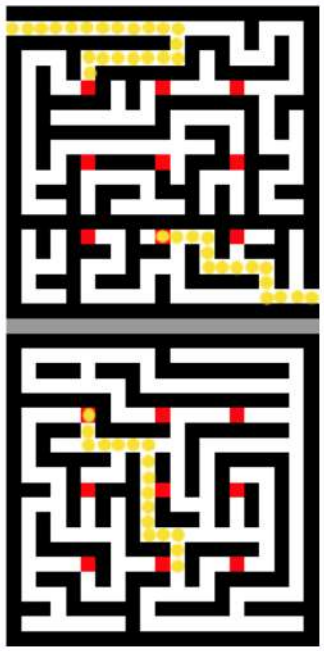
The following is a small example input data for the tunnels. You start at the entrance at top-left and need to make your way to the exit at the bottom-right. Each $ symbol represents an elevator that you can optionally take to the alternate level. Note: You can also pass through an elevator, you do not have to use it to change level.

```
###################
................#...#
##########.###.#.#.#
#...#.......#.#...#.#
#.#.#.######.#####.#
#.#..$#.#.$....$....#
#.#####.#.######.###
#.#......#.....#.#.#
#.########.###.#.#.#
#.#...........#.#...#
#.#.#$####$.###$###.#
#...#.....#.#...#...#
#.###.###.###.#.#.###
#...#.#.....#.#.#...#
###################
#.#.#$..#.$...#$..#.#
#.#.###.###.#.#.###.#
#...#...#...#.....#.#
#.###.###.#.#####.#.#
#...#.....#..........
###################

###################
#.........#.........#
#.###.###.########.#
#......#...........#
######.##########.#
#....$#...$....$....#
#.###.#####.######.#
#.#.........#......#
#.#####.#.#.#.#####.#
#...#...#.#.#.#...#.#
###.#$###.$.#.#$#.#.#
#...#.#.#.#...#.#.#.#
#.#.#.#.#.###.#.#.#.#
#.#.....#.........#.#
#.#####.###.#####.#.#
#....$#...$.#..$..#.#
#.###.#.###.#.#####.#
#...#...#.....#.....#
###.#####.#####.#####
#......#...........#
###################
```

Day 9 Part 2

The quickest path through the above example can be represented pictorially as this...



In this example, the shortest path out takes `52` steps. That would be your answer.

Try a walk through

Michael McLeod from Pakuranga College has created an interactive version of the example and final maze in javascript for you to walk through at with a first-person viewpoint! These are really cool, have a try at...

- https://haque.rs/maze/

Your task

For the map in the input data, what is the number of steps for the shortest path out of the maze?